Chapter 5

Temporal Processing with Composition-Selected Devices: Designed Filters and Reservoirs

5.1 Introduction and Scope

Chapter 4 showed that composition sets the fading-memory time of these devices over about one decade of seconds. This chapter asks what that time constant can be used for. Behavioural models fitted to the measured response are tested in two roles. In the first, one composition, or a pair of compositions, is chosen for its time constant and used as a designed filter; the references are the conventional components for the same job, an RC element with the same half-life and a digital filter. In the second, many devices form a reservoir: a bank of randomly masked nodes whose states feed a trained linear read-out. Banks drawn from one composition are compared with banks spanning several, and linear exponential-moving-

average (EMA) banks at the same half-lives serve as controls.

Peripheral physiological signals are the test bed because their timescales overlap those of the devices without rescaling: phasic electrodermal responses, respiration, and heart-rate variability vary over seconds, and tonic levels drift over minutes. The chapter uses synthetic electrodermal signals with known responses, the Wearable Stress and Affect Detection corpus (WESAD), and an independent PhysioNet corpus. Affective computing supplies the tasks and labels; the chapter does not propose a new affect classifier.

All results in this chapter are *in silico*; no new devices were fabricated. Each node is a compact behavioural model parameterised by the fits in chapter 4 (section 5.2), and every read-out is a single linear ridge regression.

5.2 Measured Behavioural Model

Each composition cell is reduced to a compact behavioural model and used as one reservoir node. The model is fitted only to the three common measurements of chapter 4: hysteresis, pulsed potentiation, and delayed depotentiation. Its parameters therefore derive from measured device data.

Each replicated composition cell is summarised by a *parameter card*: the model-free half-enhancement time $t_{1/2,c}$, any identified stretch exponent β_c, the potentiation growth exponent α_c, the measured peak ratio $R_{\mathrm{peak},c}$, and the pulse count $N_{\mathrm{peak},c}$ at that peak. The fading memory is the stretched-exponential (Kohlrausch) retention identified in chapter 4,

$$\lambda_c(\Delta t) \;=\; \exp\Big[-(\Delta t/\tau_c)^{\beta_c}\Big],$$

where $\tau_c = t_{1/2,c}/(\ln 2)^{1/\beta_c}$ is derived so that every card reproduces the primary, model-free cell-median half-time exactly. For cells without an identified stretch exponent, $\beta_c = 1$ and therefore $\tau_c = t_{1/2,c}/\ln 2$. The write envelope is

$$\frac{G(N)}{G_0} = 1 + (R_{\mathrm{peak},c} - 1)\left(\frac{N}{N_{\mathrm{peak},c}}\right)^{\alpha_c}, \qquad 0 \le N \le N_{\mathrm{peak},c}.$$

It uses the measured baseline, exponent, peak, and pulse count rather than an unbounded power law. Where a measured train turns over after its peak, the node is capped at the peak: the post-peak recovery dynamics were not measured and are not extrapolated.

Driven by a rate-coded input $u_n \in [0, 1]$ sampled at interval Δt, a device node accumulates pulse progress p and relaxes along its measured kernel:

$$p_n^{(i)} = \mathrm{clip}\left[\sum_{k\ge 0} g_k^{(i)} \frac{\mathrm{clip}(w_i u_{n-k}, 0, 1)}{N_{\mathrm{peak},i}},\ 0,\ 1\right], \qquad g_k^{(i)} = \lambda_i (k\,\Delta t)^{1/\alpha_i},$$

$$x_n^{(i)} = (p_n^{(i)})^{\alpha_i}, \qquad \frac{G_n^{(i)}}{G_0^{(i)}} = 1 + (R_{\mathrm{peak},i} - 1)\, x_n^{(i)}.$$

Here $x_i \in [0, 1]$ is the normalised enhancement, w_i is a non-negative input mask, and one full-scale sample advances the pulse-progress coordinate by at most one effective pulse. That sample-to-pulse mapping is a modelling convention, not a measured circuit timing relation. The kernel $g = \lambda^{1/\alpha}$ makes the enhancement left by a single write decay exactly as the measured λ_i, and with no relaxation and unit input the node follows the measured envelope to its peak and cannot exceed it. Because the sum runs over the whole measured relaxation, a stretched decay is reproduced at any sampling interval. Applying the one-step factor $\lambda_i(\Delta t)$ repeatedly would instead compound into a plain exponential and, for $\beta < 1$, shorten the simulated memory (for the lead cell at $\Delta t = 1\,\mathrm{s}$, to a half-life of 11.5 s instead of the measured 19.2 s). A node that reaches its peak is held

there, with its stored history rescaled (Appendix D). The internal state is assumed to be read sub-threshold without disturbance, as in the proof-of-concept device of chapter 2. Every read-out is a single *linear* ridge regression on the device states.

Write nonlinearity and fading memory were measured at different write amplitudes (3 V for potentiation and 4 V for depotentiation; table 4.1). Combining them in one node assumes that the two regimes are compatible and probably underestimates the coupling between drive and retention. Both measurements also used a single, effectively common inter-pulse cadence, so the model does not resolve input timing below that cadence. The gold/TMPE negative-polarity corpus of chapter 4 measures integration and relaxation within single runs and finds both interval-dependent potentiation and written-depth-dependent relaxation. Because it uses another host, electrode, polarity, and read protocol, it does not provide parameters for the banks used here. The experiment needed for the lead composition is specified in section 5.12. Per-cell parameter cards and all simulation and read-out settings are tabulated in Appendix D.

Component cards. Read as circuit elements, the cards form the component table of table 5.1. Each cell is a low-pass element whose time constant is set by composition, whose corner frequency follows from its measured kernel, and whose tolerance is the within-cell scatter. For a single-exponential cell the corner is $f_c = 1/(2\pi\tau_{RC})$. For a stretched cell it lies lower, because the long tail of the relaxation retains low frequencies, so the half-life alone does not fix the filter.

Table 5.1: Component cards of the replicated PEO/LiOTf grid, ordered by half-life. $t_{1/2}$ and β define the measured relaxation; $\tau_{RC} = t_{1/2}/\ln 2$ is the time constant of the single-exponential (RC) element with the same half-life; f_c is the $-3\,\mathrm{dB}$ corner of the low-pass formed by the measured kernel and by that RC element; α, R_{peak}, and N_{peak} describe the pulse response. n counts fabricated substrates. The median within-cell tolerance is $\sigma(\ln t_{1/2}) = 0.26$ (cell range 0.09–1.02; Appendix D).

Cell	n	$t_{1/2}$ (s)	β	τ_{RC} (s)	f_c device (Hz)	f_c RC (Hz)	α	R_{peak}	N_{peak}
PEO 1.2/0.045	3	2.7	1.00	4.0	0.0404	0.0404	0.33	4	100
PEO 0.6/0.045	2	2.8	0.97	4.0	0.0375	0.0401	0.57	15	100
PEO 0.6/0.09	3	3.5	0.61	5.0	0.0085	0.0316	0.83	202	300
PEO 0.6/0.18	2	5.5	0.61	8.0	0.0054	0.0200	0.63	77	1000
PEO 1.2/0.09	3	6.0	0.91	8.6	0.0152	0.0185	0.49	43	1000
PEO 1.2/0.18	3	9.0	0.87	13.0	0.0092	0.0122	0.37	8	1000
PEO 0.3/0.045	4	15.8	0.64	22.8	0.0022	0.0070	1.11	483	300
PEO 0.3/0.09 (lead)	3	19.2	0.83	27.7	0.0038	0.0058	0.90	150	1000

5.3 Designed Filters from the Component Cards

Before any reservoir is assembled, each composition cell can be used on its own. Its relaxation kernel defines a causal low-pass filter whose corner frequency is fixed when the film is formulated (table 5.1). Two elementary circuits follow from it. A device subtracted from its own input forms a high-pass baseline remover, $y = x - \mathrm{LP}_c(x)$. Two compositions subtracted from each other form a band-pass, $y = \mathrm{LP}_{\mathrm{fast}}(x) - \mathrm{LP}_{\mathrm{slow}}(x)$, which gives no output for a constant input. Both are routine preprocessing steps for slow sensor signals.

Electrodermal activity (EDA) is the test case. Its phasic part, the skin-conductance responses (SCRs) that follow sympathetic arousal, rides on a tonic level that drifts over minutes. Conventional pipelines remove the tonic level with a high-pass filter near 0.05 Hz and then detect SCRs or measure power in the 0.045 Hz–0.25 Hz band associated with sympathetic tone [1]. Three references frame the comparison: an RC element with the same half-life as the device, which is the conventional component for

the same job; a causal digital Butterworth phasic filter (0.05 Hz–1 Hz); and the unfiltered signal.

Each device is represented by its measured kernel $\exp[-(t/\tau_c)^{\beta_c}]$, applied as a causal convolution at 4 Hz on the assumption that the conductance enhancement superposes. Every simulated device draws its own time constant from the measured within-cell scatter, $\sigma(\ln t_{1/2}) = 0.264$; the *RC* references are ideal. The composed nonlinear node of section 5.2 gives the same ranking once its read-out gain is set for small signals (Appendix D).

Design space. All eight high-pass corners lie between 0.003 and 0.041 Hz, below the lower edge of the sympathetic band (fig. 5.1a). The two fastest cells ($t_{1/2} \approx 2.7$ s) place the corner close to the conventional 0.05 Hz tonic cut-off. No cell is fast enough to set the upper edge of the band. A two-composition band-pass therefore passes 0.002 Hz–0.047 Hz, which is the tonic range. For stretched cells, the low-pass corner also sits well below that of the *RC* element with the same half-life (PEO 0.6/0.09: 0.009 against 0.032 Hz), because the long tail of a stretched relaxation retains low frequencies. The half-life alone therefore does not specify a filter; its corner has to be read from the measured kernel, as in table 5.1.

Synthetic signals with known responses. Synthetic sessions supply ground truth. They contain Bateman-shaped SCRs [2] at about two and seven per minute in alternating five-minute calm and aroused blocks, a tonic rise in the aroused blocks, minutes-scale drift, and either low noise, broadband noise, or sharp motion-like artefacts. Detection thresholds and lags are calibrated on 12 sessions and scored on 30 others (fig. 5.1b,c). The fastest single-device high-pass detects SCRs with F1 0.713, 0.680, and 0.676 in the three conditions, against 0.701, 0.669, and 0.647 for the digital filter; under artefacts it is higher in 29 of 30 sessions. The *RC* element with the same half-life scores 0.712, 0.677, and 0.675. The best

two-composition band-pass reaches only 0.50–0.54 and passes about twice as much tonic as phasic power. The unfiltered signal scores 0.690 on clean data, because peak prominence already discounts a slow baseline; filtering helps mainly against noise and artefacts.

WESAD recordings. On WESAD, per-minute phasic power is compared between the stress and baseline segments of each subject (fig. 5.1d). The single-device high-pass gives a within-subject AUC of 0.952 on the chest and 0.921 on the wrist, against 0.951 and 0.927 for the digital filter and 0.947 and 0.922 for a filter restricted to the sympathetic band. The per-minute features of the device and digital filters are closely correlated (Spearman $\rho = 0.98$ and 0.99). The tonic level alone gives 0.915 and 0.885. The device-minus-digital difference is +0.001 on the chest ($p = 0.94$) and −0.006 on the wrist (2/15 subjects higher, $p = 0.038$). The two-composition band-pass is lower at both sites (0.903 and 0.880).

Device against *RC* element. Across all eight cells, the measured kernel never improves on the *RC* element with the same half-life (Appendix D). The fastest cell relaxes as a simple exponential ($\beta = 1.00$) and is therefore the same filter as its *RC* counterpart. The two most stretched cells ($\beta = 0.61$) are slightly worse on the chest (−0.009 AUC; $p = 0.013$ and 0.038). Used as a filter, a composition-selected device is an *RC* element whose time constant is set at fabrication. Its case rests on how that time constant is obtained. A $0.0825\,\mathrm{cm}^2$ two-terminal film reaches seconds with no separate capacitor, and impedance spectroscopy places the storage in the ionic response of the film (section 4.7). In analogue CMOS, time constants of hundreds of milliseconds already need large capacitors or very small currents [3], and printed circuits have used printed low-pass filters for temporal processing, including printed EDA stress detection [4, 5]. Footprint, voltage, stability, and tolerance were not

compared against a printed *RC* element realising the same time constant; that comparison would decide the practical value of the device.

5.4 Reservoir Computing and the Common Metric Language

The reservoir principle. The remaining sections test the second use of the cards, as nodes of a reservoir. Reservoir computing uses a fixed dynamical system whose state is connected to a trained linear read-out [6–8]. A physical reservoir requires fading memory, so its state retains recent input history, and a nonlinear response that maps inputs into features accessible to the read-out. Chapter 4 measures both properties in these devices: stretched-exponential depotentiation supplies fading memory, and compressive potentiation supplies nonlinearity. Memristive and organic implementations have previously used the same combination [9–12]. The behavioural models test whether the measured polymer-electrolyte response supplies that combination.

Architecture. The reservoir is a spatial bank of N *independent* device nodes, each represented by the behavioural model of section 5.2 and driven by a randomly masked copy of the input. With no inter-node recurrence, differences between banks arise from their intrinsic timescales, device-to-device scatter, and input masks rather than engineered connectivity. Each reservoir result is paired with a linear control: exponential moving averages (EMAs) of the input channels at the device half-lives, $\tau = t_{1/2}/\ln 2$, read out in the same way. The EMA bank has the time constants of the device bank but none of its nonlinearity or random masking. For Demonstration A, the same single-composition node could instead be deployed as a time-multiplexed delay-feedback reservoir with virtual nodes

along a delay line [13]; the spatial bank is used throughout for consistency.

Common metrics. Linear memory capacity MC_k and its total [14] quantify the past input recoverable at each lag. NARMA-10 [15] provides a nonlinear system-identification benchmark, and the information-processing-capacity framework [16] separates linear and nonlinear contributions (fig. 5.4). Affective tasks are scored by class-balanced macro-F1. Robustness is tested by injecting the measured within-cell timescale spread into the bank (fig. 5.9); no common scatter estimate is assumed for α or β.

5.5 Benchmark Tier: Capacity and System Identification

Reservoir suitability is first evaluated without an application-specific task. Linear memory capacity measures fading memory, while information-processing capacity separates its linear and nonlinear components. The NARMA-10 system-identification task is used only to rank compositions, not to claim absolute performance. All three benchmarks use a random rate-coded input, N device nodes, and a linear ridge read-out.

Statistical units. Random input and bank seeds sample the algorithm's sensitivity to stochastic initialisation; they are not independent devices, experiments, or subjects. Seed-only benchmark comparisons therefore report means, standard deviations, and sign consistency without inferential p-values. Inferential paired tests are reserved for independent held-out subjects in the physiological-corpus analyses, or for independently fabricated devices in the experimental chapters. The complete distinction is recorded in Appendix D.

Memory capacity. The linear memory capacity of Jaeger measures how much of the past input a reservoir can reconstruct: MC_k is the squared correlation between a ridge reconstruction of the input k steps ago and its true value, and the total is the sum over all lags, bounded above by the number of nodes N [14]. Figure 5.2 compares two banks of equal size ($N = 24$): a *homogeneous* bank of jittered copies of the lead cell (PEO 0.3/0.09) and a *heterogeneous* bank spanning the replicated composition grid ($\tau \approx 4$–$30\,\mathrm{s}$). Total capacity rises from 3.22 ± 0.32 to 4.44 ± 0.45 (means $\pm$ SD over ten random input/bank seeds), a ratio of 1.38 ± 0.14 and a paired gain of 1.22 ± 0.41 that is positive in all ten seed pairs. Most of the gain sits at the shortest lags: single-step recall rises from $\mathrm{MC}_1 = 0.41$ to 0.98, whereas recall at lag 15 is similar (0.08 and 0.07). Copies of one slow cell carry strongly correlated states, and the faster cells of the heterogeneous bank supply the short-lag information they lack.

Composition sweep. The same benchmarks, run on single-composition banks, rank the compositions against one another (fig. 5.3). The strongly stretched PEO 0.6/0.09 cell ($\beta = 0.61$) attains the highest single-composition memory capacity (3.35 ± 0.36, mean $\pm$ SD over ten seeds) and the lowest error on the NARMA-10 system-identification benchmark [15] (NRMSE 0.614 ± 0.018); the lead cell PEO 0.3/0.09 gives 2.96 ± 0.26 and 0.692 ± 0.033. A stretched relaxation is a distribution of time constants, so one stretched device already spreads its memory over several lags. The differences between neighbouring cells are algorithmic rankings, not experimental evidence of a unique optimum. The NARMA-10 numbers carry only *relative* weight: the absolute error is high (≈ 0.6–0.9) because a bank of independent, non-recurrent nodes is a weak NARMA-10 reservoir.

Comparison with randomly distributed timescales. A generic echo-state bank of the same size tests whether the measured timescales are individually important or whether their range is sufficient. A random bank spanning approximately the same range ($\tau = 3$–$30\,\mathrm{s}$) has a memory capacity of 4.97 ± 0.78, compared with 4.44 ± 0.45 for the device bank (ten matched seeds; the device is higher in four). Expanding the random range to 0.5–60 s raises the capacity to 5.56 ± 0.74 (the device is higher in one). The measured values are therefore not uniquely optimal; this control attributes the available linear memory to the range of fading times rather than to their exact distribution. The broader-range result is only a modelling prompt: the corresponding slow devices have not been measured.

Linear and nonlinear capacity. Memory capacity counts only linearly recoverable information. The information-processing-capacity framework of Dambre et al. decomposes total capacity into orthogonal polynomial degrees: degree-one terms give linear capacity, while degree-two terms measure nonlinear self and cross-lag transformations; the total is bounded by the node count N [16]. Figure 5.4 reports this decomposition. The homogeneous bank has total capacity 5.18 ± 0.34, comprising 3.61 linear and 1.57 nonlinear capacity. The heterogeneous bank reaches 9.77 ± 0.73, comprising 5.06 linear and 4.71 nonlinear capacity; its nonlinear increment of 3.13 is positive in all ten seed pairs. Of the heterogeneous nonlinear capacity, 2.55 is from self terms and 2.16 from cross-lag products. Like the linear comparison, this is a sensitivity result for the composed behavioural model rather than an experimental population estimate.

Table 5.2: Characteristic timescales of the peripheral signals used in this chapter, and the composition cells whose fading-memory window covers them [18–20]. The grid of chapter 4 covers the phasic, respiration, and heart-rate-variability bands; no measured cell reaches the minutes-scale tonic band.

Affective signal	Timescale	Device coverage
Phasic EDA (SCR)	∼1–10 s	low–mid PEO ($t_{1/2}$ 3–19 s)
Respiration	∼3–5 s	mid PEO
HRV, HF band	∼2.5–7 s	mid PEO
HRV, LF band	∼7–25 s	PEO 0.3 ($t_{1/2} \approx$ 16–19 s)
Tonic SCL	minutes	not reached by the measured grid

5.6 Physiological Signals as a Test Bed

Affective computing uses physiological signals to infer emotional and physiological state [17]; here its signals and labels serve as a test bed for the device models. The device timescales, from a few seconds to tens of seconds, overlap the phasic electrodermal, respiration, and heart-rate-variability bands without time rescaling (table 5.2). These signals also contain several timescales at once: phasic electrodermal responses, respiration and heart-rate-variability rhythms over seconds, and tonic levels drifting over minutes [18–20]. Figure 5.5 places the measured relaxation time of each composition on the same axis as these physiological bands and shows the minutes-scale range not reached by composition alone.

Extending the model to tonic timescales. Raising the write amplitude lengthens fading memory in the measurements of chapter 4. Adding modelled nodes at $\tau = 120$ and 300 s changes little: random-input capacity beyond 75 s rises from 0.82 to 0.97 (9/10 seed pairs), total capacity is unchanged (4.67 and 4.64), and slow-band physiological reconstruction at delays up to 120 s rises by only $+0.003 \pm 0.022$ R^2 (5/10 draws; Ap-

pendix D). The stretched tails of the measured cells already carry much of this long-lag memory. The extension is hypothetical in any case, because no minutes-scale cell was measured and the drive-amplitude data do not identify these two time constants.

The experiments that follow use the Wearable Stress and Affect Detection corpus (WESAD) of Schmidt et al. [21]: fifteen subjects wearing a chest sensor that records electrodermal activity, respiration, temperature, and an electrocardiogram, with periods labelled baseline, stress, or amusement. WESAD is used in two complementary ways: as a source of real physiological streams whose multi-lag temporal context the reservoir must encode (section 5.7), and as a downstream affective-classification task (section 5.8).

5.7 Physiological Temporal-Context Reconstruction

The benchmarks of section 5.5 show that a spread of device timescales broadens memory on a random input. This section tests the same property on the multi-timescale physiological signals introduced in section 5.6, isolating temporal-context encoding before turning to affect labels (section 5.8).

The reservoir is run continuously over each WESAD session at $\Delta t = 1\,\mathrm{s}$ on four chest channels (electrodermal activity, respiration, temperature, and an electrocardiogram-derived heart rate), and a linear read-out reconstructs each channel at delays of 1, 3, 8, 20 and 45 s. These delays span fast phasic and beat-to-beat context, respiration- and HF-HRV-scale context, and tonic, LF-HRV-scale context [18, 19]. Evaluation is leave-one-subject-out over all fifteen subjects, with $N = 48$ nodes per bank.

Preprocessing is causal. ECG filtering and RR-interval assignment use only completed beats; resampling uses the nearest past label. Each channel is mapped to $(0, 1)$ by a monotone transform fitted to the first 60 s only: the calibration-prefix 5th and 95th percentiles map to 0.25 and 0.75, without hard clipping later values. Thus neither later held-out values nor affect labels set the scale. The reported out-of-sample R^2 uses the training-subject mean as its deployable null predictor, clips reconstructed normalised physiology to $[0, 1]$, and weights each held-out subject equally.

Table 5.3 reports the held-out reconstruction quality. A memoryless bank gives $R^2 = 0.636$, compared with 0.646 for instantaneous input. A single-timescale device bank raises R^2 to 0.679–0.684, and the heterogeneous composition bank reaches 0.707: +0.061 over instantaneous input and +0.023 over the stronger same-size homogeneous control. The heterogeneous-minus-fast and heterogeneous-minus-slow differences are each positive in 14/15 subjects (Holm-adjusted $p = 0.0067$). The fast homogeneous bank is strongest at the shortest delays and the slow homogeneous bank at the longest (fig. 5.6). A linear EMA bank at the eight cell half-lives, with no device nonlinearity or random mask, reaches 0.742 and exceeds the heterogeneous device bank in all fifteen subjects (-0.035; Holm-adjusted $p = 1.8 \times 10^{-4}$); a single EMA at the lead half-life gives 0.688. The useful content of the bank is therefore its set of time constants, and the nonlinear write and random masks lower the reconstruction.

5.8 WESAD Affective Classification

The downstream task is affective-state classification from the WESAD physiological streams with a leave-one-subject-out linear read-out. The reservoir and the 15 s causal prediction smoother run on the complete physical timeline; samples with undefined affect labels are excluded only from fitting and scoring. This prevents the smoother from joining labelled

Table 5.3: Subject-mean leave-one-subject-out R^2 for multi-lag physiological context reconstruction on WESAD ($N = 48$ nodes; means over ten bank seeds where applicable). The null prediction is the training-subject mean. Bands are reconstruction delays: fast = 1–3 s, mid = 8 s, slow = 20–45 s. The heterogeneous bank exceeds the stronger homogeneous control by +0.023 R^2 (14/15 subjects; Holm-adjusted $p = 0.0067$), and the linear EMA bank at the same half-lives exceeds the heterogeneous bank by 0.035 (15/15 subjects).

Bank	overall R^2	fast	mid	slow
Instantaneous	0.646	0.699	0.647	0.592
Memoryless	0.636	0.691	0.637	0.580
Homogeneous, fast	0.679	0.777	0.655	0.593
Homogeneous, slow	0.684	0.687	0.694	0.677
Heterogeneous	0.707	0.757	0.693	0.664
EMA, lead half-life	0.688	0.692	0.695	0.682
EMA bank, 8 half-lives	**0.742**	**0.794**	**0.724**	**0.699**

blocks separated by long protocol gaps. The study that introduced WESAD reports, with engineered window features and conventional classifiers under leave-one-subject-out validation, chest macro-F1 of 0.725–0.744 for the three classes and 0.904–0.915 for stress versus non-stress, and 0.861 for stress versus non-stress from the wrist [21]. The streaming linear read-out of this chapter reaches 0.562, 0.799, and 0.756 on the same three tasks (table 5.4). The formulations differ (per-step streaming against windowed features), but a gap of 0.1–0.2 is too large to attribute to formulation alone: the device models do not approach an engineered-feature pipeline on these labels.

Demonstration A: window-level electrodermal classification. Binary stress-versus-baseline classification from electrodermal activity, read out at the window level from the lead composition alone, reaches leave-one-subject-out macro-F1 0.671 (fig. 5.7a). A static baseline using the mean, standard deviation, last value, and slope of the same signal reaches

Table 5.4: Matched comparison with the study that introduced WESAD. All rows report class-balanced macro-F1 under leave-one-subject-out validation over the same fifteen subjects; binary rows separate stress from baseline and amusement pooled. Schmidt et al. give the two best classifiers for each task. This work reports the heterogeneous device bank (three-class: seed mean; binary: clean streams). Accuracy-only and differently split studies are omitted.

Task	Study	Features and read-out	macro-F1
Three-class, chest	Schmidt et al. [21]	window features; AdaBoost, LDA	0.725, 0.744
	This work	streaming device states; linear ridge	0.562
Binary, chest	Schmidt et al. [21]	window features; RF, LDA	0.904, 0.915
	This work	streaming device states; linear ridge	0.799
Binary, wrist	Schmidt et al. [21]	window features; AdaBoost, RF	0.859, 0.861
	This work	streaming device states; linear ridge	0.756

0.714. The reservoir does not improve this 60 s-window task.

Demonstration B: streaming three-class tracking. The reservoir runs continuously over each session, and baseline, stress, or amusement is read out at every labelled step. Table 5.5 decomposes the result into four matched-size banks. A majority-class predictor reaches macro-F1 0.231 and a stratified-random predictor 0.333. Instantaneous input reaches 0.482; the memoryless bank reaches 0.513 ± 0.038; and the homogeneous fading-memory bank reaches 0.562 ± 0.038. The heterogeneous bank also reaches 0.562 ± 0.051, a $+0.080$ gain over instantaneous input and $+0.001 \pm 0.046$ relative to homogeneous, with heterogeneity higher in four of ten matched seed pairs. Linear EMA controls at the lead half-life and at all eight half-lives score 0.503 and 0.508. At the representative seed,

Table 5.5: Leave-one-subject-out macro-F1 for streaming three-class WESAD tracking (means ± SD over ten matched bank seeds; the linear controls have no seed dependence). Gains are relative to instantaneous input. The two device banks tie and both exceed linear EMA controls at the same half-lives; adding the measured timescale spread does not improve the sustained-label task.

Features	macro-F1	gain
Instantaneous (no memory)	0.482	—
Memoryless bank	0.513 ± 0.038	+0.031
Homogeneous bank (one τ)	$\mathbf{0.562 \pm 0.038}$	+0.080
Heterogeneous bank (τ spread)	$\mathbf{0.562 \pm 0.051}$	+0.080
Linear EMA, lead half-life	0.503	+0.021
Linear EMA bank, 8 half-lives	0.508	+0.026

amusement remains the weakest class (F1 0.40, versus 0.83 for baseline and 0.77 for stress). The experiment supports a memory benefit relative to an instantaneous read, and on this task the device banks also exceed the linear filters, but it shows no timescale-diversity benefit for these sustained labels.

Why heterogeneity does not improve the label scores. The affect labels are sustained states that change only at protocol boundaries, so one slow timescale already carries most of the history that separates them. A wider timescale distribution helps when the target explicitly requires several lags (table 5.3), but not here. This is also the one task in the chapter where the device banks exceed linear filters at the same time constants: 24 device states score 0.562, whereas the 32 features of the EMA bank score 0.508.

5.9 Continuous Monitoring Under Deployment Conditions

The sustained WESAD states can often be classified from the instantaneous signal level, as shown by the static-feature baseline. Wearable measurements also contain sensor noise and motion artefacts, particularly in wrist-worn electrodermal and photoplethysmographic signals. Fading memory implements temporal low-pass filtering and may therefore reduce sensitivity to transient artefacts. The tests below add synthetic noise, measured-motion gating, and an independent sensor corpus.

Figure 5.8 tests a continuous binary stress detector (stress versus baseline plus amusement, read out at every step, leave-one-subject-out, with the same linear read-out used throughout). On clean streams, macro-F1 ranges from 0.781 to 0.807. At $\sigma = 0.4$, instantaneous and memoryless reads fall to 0.641 and 0.649, while the homogeneous and heterogeneous memory banks retain 0.818 and 0.811. The heterogeneous-minus-instantaneous subject difference is +0.171 (12/15 positive, paired Wilcoxon $p = 0.015$). Clean false-alarm rates are similar (instantaneous 0.094, heterogeneous 0.089 seed-mean). Transition-window F1 changes from 0.832 to 0.840, but the subject difference is not resolved (6/15 positive, $p = 0.58$); all banks have zero median onset latency under the five-second stable-call definition. Temporal smoothing is useful under corruption, not for earlier detection in this protocol.

Accelerometer-gated motion artefact. Because the preceding corruption is synthetic, a second test gates its amplitude with the measured chest accelerometer. At the strongest gating, the instantaneous read falls to 0.702, the heterogeneous bank retains 0.811, and the EMA reaches 0.809. Both gains over the instantaneous read are +0.111; the EMA gain is resolved ($p = 0.035$) and the heterogeneous gain is not ($p = 0.055$;

Appendix D). The data-driven control therefore supports temporal filtering under measured-motion corruption, but not a separate device-model advantage.

Classical-filter control. A single exponential moving average at the lead half-life reaches macro-F1 0.804 at $\sigma = 0.4$, compared with 0.641 for instantaneous input; its paired subject gain is +0.170 ($p = 0.022$). A linear EMA bank at the eight half-lives reaches 0.808, with no resolved gain over the single EMA (+0.003, $p = 0.35$). The heterogeneous device bank reaches 0.811 and does not differ from the EMA bank (-0.002, $p = 0.64$; Appendix D). Robustness is therefore attributable to temporal low-pass filtering at these time constants, which the device bank performs as well as the linear control and no better.

Cross-corpus replication. The robustness test was repeated on the PhysioNet Non-EEG corpus of Birjandtalab et al. [22]: twenty subjects wearing an Affectiva Q wrist sensor for electrodermal activity and skin temperature, with pulse-oximeter heart rate. Cognitive and emotional stress are compared with relaxation; physical exertion is excluded. On clean streams, seed-mean heterogeneous and instantaneous scores are 0.657 and 0.639; the representative-seed subject difference is not resolved ($p = 0.11$). At $\sigma = 0.4$, they are 0.653 and 0.451: a +0.201 gain, positive in all twenty subjects ($p = 1.9 \times 10^{-6}$; table 5.6). The linear EMA bank retains 0.633, and the device bank exceeds it by 0.023 (16/20 subjects, $p = 0.017$). This independent corpus reproduces the noise-response ordering, while confirming that the clean-stream advantage is small.

Table 5.6: Noise-response comparison on the independent PhysioNet Non-EEG stress corpus [22] (twenty subjects; leave-one-subject-out, linear read-out). At $\sigma = 0.4$, the heterogeneous-minus-instantaneous difference is $+0.201$, positive in 20/20 subjects ($p = 1.9 \times 10^{-6}$). Clean-stream scores are comparable.

Detector	macro-F1 (clean)	macro-F1 ($\sigma = 0.4$)
Instantaneous (no memory)	0.639	0.451
Homogeneous (memory)	**0.678**	**0.669**
Heterogeneous (memory)	0.657	0.653
Linear EMA bank	0.647	0.633

5.10 Toward a Worn Device: Sensor Site and Operating Budget

The robustness of the previous section is a laboratory statement until two further questions are addressed: whether the advantage survives a consumer-grade sensor, and what can be said about an operating budget. The first is tested with the WESAD wrist recordings. For the second, only a reference calculation is possible because the simulated PEO nodes were not measured as a wired bank.

Sensor site. WESAD records each session with both a chest strap and an Empatica E4 wrist band. The wrist band has fewer and noisier channels: skin conductance, skin temperature, and photoplethysmography-derived heart rate, without respiration. On clean wrist data (table 5.7), instantaneous and heterogeneous scores are 0.724 and 0.756. At $\sigma = 0.4$ they are 0.615 and 0.769; the paired subject gain is $+0.117$ (10/15 positive, $p = 0.035$). The corresponding chest scores are 0.641 and 0.811. The linear EMA bank gives 0.773 clean and 0.755 at $\sigma = 0.4$ on the wrist, and the device bank does not differ from it (-0.013, $p = 0.38$). Clean wrist false-alarm rates are not improved (instantaneous 0.112, heterogeneous

Table 5.7: Continuous binary stress-detection macro-F1 on the WESAD chest strap versus the Empatica E4 wrist band (leave-one-subject-out, linear read-out; seed means). Under injected corruption, the temporal bank exceeds instantaneous input at both sites and does not differ from the linear EMA bank; σ follows fig. 5.8.

	Chest		Wrist E4	
Detector	clean	$\sigma = 0.4$	clean	$\sigma = 0.4$
Instantaneous (no memory)	0.781	0.641	0.724	0.615
Memory reservoir (het)	0.799	**0.811**	0.756	**0.769**
EMA bank, 8 half-lives (linear)	**0.806**	0.808	**0.773**	0.755

0.106), so the supported benefit is robustness under injected corruption rather than fewer clean-stream alarms.

Operating-budget reference. The parameter count is exact for the simulated architecture: the device dynamics are fixed, and the fitted $(N{+}1) \times K$ read-out contains 75 weights for $N = 24$, $K = 3$, or 147 for $N = 48$. The read-out also requires NK multiply–accumulates per step. Energy is less certain. Recalculation of one proof-of-concept *Super Yellow*/*Hybrane*/LiOTf junction gives 0.084 µJ–1.55 µJ across its positive write sequence (median 1.08 µJ). Multiplying that median by 24 nodes gives a reference core cost of 25.9 µJ per update, or 25.9 µW at one update per second; the corresponding 48-node value is 51.8 µW.

These products are not power predictions for the proposed reservoir. They transfer a 1 V result from one large-area *Hybrane* junction to PEO models identified at 3 V–4 V, without measured bank wiring, read energy, or peripheral circuits. They exclude the sensor analogue front-end, analogue-to-digital conversion, addressing, and control, and are neither an upper nor a lower bound. The device response remains faster than the sub-hertz physiological signals considered here, and its relaxation would perform the temporal filtering during the write/read cycle. The assumptions are

tabulated in Appendix D.

5.11 Matching Device and Task Timescales

Choosing a cell. For a designed filter, the target corner frequency selects the composition (table 5.1). The two fastest cells, with corners near 0.04 Hz, are the baseline removers for electrodermal activity; slower cells smooth over tens of seconds and pass the tonic band. Because stretching lowers the corner, a stretched cell must be chosen by its corner frequency, not by its half-life. Corners above about 0.04 Hz lie outside the replicated grid; the faster host and anion variants of chapter 4 (TMPE, $t_{1/2} \approx 4\,\mathrm{s}$; TFSI, $\approx 1\,\mathrm{s}$) indicate where faster elements might come from, at survey-level evidence.

Reservoir banks. Performance depends on the overlap between device and task timescales (fig. 5.5). In the single-composition sweep, the strongly stretched PEO 0.6/0.09 cell gives the highest memory capacity for the seconds-scale benchmark (fig. 5.3). For the multi-lag physiological task, no single cell covers the full delay range, and the heterogeneous bank gives the highest device-bank score, below the linear EMA bank at the same half-lives (table 5.3).

Effect of timescale breadth. Table 5.8 separates fading memory from timescale breadth and from the device nonlinearity. On random input, breadth raises total linear memory ($+1.22 \pm 0.41$, positive in 10/10 seeds) and nonlinear information-processing capacity. On real physiology, heterogeneity gives a +0.023 context-reconstruction gain over the stronger homogeneous control, but a linear EMA bank at the same half-lives does better than either. On sustained three-class labels the two device banks

Table 5.8: Comparison of instantaneous input, homogeneous and heterogeneous device banks, and a linear EMA bank at the eight cell half-lives across the physiological tasks in this chapter. Temporal features show their largest gains under corruption. The linear bank gives the highest physiological-context score and the device banks the highest three-class score; under corruption the device and linear banks are close. Bold marks the highest score per row.

Task	Metric	Instant.	Homog.	Heterog.	EMA bank
Physiological context (multi-τ)	R^2	0.646	0.684	0.707	**0.742**
Three-class affect tracking	macro-F1	0.482	**0.562**	**0.562**	0.508
Stress monitoring, clean	macro-F1	0.781	**0.807**	0.799	0.806
Stress monitoring, σ=0.4	macro-F1	0.641	**0.818**	0.811	0.808
Wrist stress, clean	macro-F1	0.724	0.752	0.756	**0.773**
Wrist stress, σ=0.4	macro-F1	0.615	0.766	**0.769**	0.755

tie, and under corruption the homogeneous bank is marginally stronger. The task, not diversity alone, determines the useful distribution.

When one slow timescale dominates, as in the sustained stress labels and the stress-monitoring task, the homogeneous and heterogeneous banks are within 0.008 macro-F1 of each other at every site and noise level. The available variation comes from composition and device-to-device scatter; chemistry and drive amplitude provide additional, sample-limited axes in chapter 4.

Operating point and integration. The models apply in the measured sub-kilohertz regime, which covers the physiological signals used here but not fast logic. The assumed sub-threshold, state-preserving read sets the read-voltage and read-disturb limits for a circuit implementation. Figure 5.9 varies timescale-only jitter. Its primary value, $\sigma(\ln t_{1/2}) \approx 0.264$, is the median within-cell substrate-level spread on the screened

replicated grid (Appendix D). The observed per-cell estimates span 0.089–1.018, so the wider sweep is a sensitivity analysis rather than a fitted uncertainty distribution. The legacy single-exponential fit gives a larger $\sigma(\ln \tau) \approx 0.85$, retained only as an upper stress-test value. No measured common jitter is applied to α or β. At zero timescale scatter, copies of one card are more redundant; the heterogeneous bank still differs through composition-dependent card parameters.

5.12 Limitations and Scope Conditions

In silico. The demonstrations are simulations. Each node is the measured $\phi \otimes \lambda \otimes f$ behavioural model of section 5.2, not a bench circuit. Every figure caption names the data behind it. Neither a wired filter nor a wired reservoir was built.

Designed filters. The filter results assume that the conductance enhancement superposes and that the measured large-signal decay also governs small signals around a written state. The composed node predicts a faster small-signal kernel when $\alpha < 1$ (Appendix D); a small write superposed on a written state would decide between the two. The synthetic generator uses chosen parameter ranges rather than a fitted corpus, and the WESAD stress segment (a speaking task) may add movement-related phasic power; this affects all filters alike. The *RC* references are ideal components, and footprint, energy, and tolerance were not compared against a fabricated *RC* element.

The composition assumption. As detailed in section 5.2, write nonlinearity and fading memory were measured separately, at different write amplitudes and one inter-pulse cadence. Combining them assumes that the two regimes are compatible; even the direction and magnitude of the

omitted coupling are uncertain for the silver/PEO system. In a sensitivity analysis, each node's retention was rescaled with its write exponent over $\kappa = -0.5$ to $+0.5$. The mean heterogeneous-to-homogeneous capacity ratio stays between 1.29 and 1.42, with the heterogeneous bank higher in every seed pair (Appendix D). The gold/TMPE negative-polarity measurements provide an external directional reference, $\rho = +0.46$ between write depth and decay half-time across 28 single-run traces. The nominal cards already have a stronger across-card rank association ($\rho \approx +0.57$), so the central external estimate maps to $\kappa^\star = 0$ and its upper confidence limit to about $+0.31$; the capacity ratio is 1.38 and 1.33 at these points. Because host, electrode, polarity, and protocol differ, this is a sensitivity anchor rather than validation of the composed silver model. A matched-amplitude experiment on the lead composition that varies pulse count and inter-pulse interval together is required.

Operating regime and read-out. The model is identified only in the measured sub-kilohertz regime; fast temporal tasks are outside its scope. Every read-out is a single ridge regression because a nonlinear read-out could supply the task nonlinearity independently of the reservoir. The read is assumed to be sub-threshold and state-preserving.

Heterogeneity axes. The quantitative comparisons span the replicated PEO/LiOTf composition grid; the single recovered PEO 0.15 substrate is excluded. Chemistry and drive amplitude enter only as secondary controls because their matched cells contain at most two substrates (chapter 4); Demonstration B is a proof of principle for composition heterogeneity, not a powered comparison across chemistries. The cards index composition, but thickness and other fabrication variables are not fully separable from that design. The available covariate checks do not explain the composition association, yet they cannot prove composition is the sole cause (fig. 4.1,

section 4.2).

Corpora, protocol, and negative controls. The affective results use WESAD as the main corpus, and noise robustness is replicated on the PhysioNet Non-EEG dataset (section 5.9). A broader multi-corpus study remains future work. In WESAD, the experimental blocks follow a fixed protocol rather than a counterbalanced order. Affect label, elapsed time, and slow session drift are therefore partly entangled: causal prefix calibration prevents future-sample leakage, but it cannot establish that a classifier is specific to affect rather than protocol position. The sustained block labels also do not provide independent ground truth for physiological onset, so the transition and latency analyses quantify agreement with protocol boundaries, not the onset of an internal state.

The strongest robustness results use synthetic, channel-wise corruption and should be read as controlled stress tests. The measured-motion gate is a more realistic but weaker control and does not resolve a heterogeneous-over-instantaneous advantage. The linear controls bound the device-specific contribution. A linear EMA bank at the same half-lives exceeds the device bank on physiological reconstruction and matches it under synthetic corruption on the chest and wrist (Appendix D); the device banks exceed the linear controls only on the sustained three-class labels and, by 0.023, under corruption on the Non-EEG corpus. Timescale heterogeneity does not improve the final-label score (section 5.8).

5.13 Summary

Read as component cards, the replicated compositions supply time constants of about 4 to 28 s and low-pass corners of 0.002 Hz–0.04 Hz. Used as a designed filter, the fastest cell removes the tonic level of electrodermal

signals as well as a digital filter (synthetic SCR detection F1 0.713 against 0.701; WESAD chest stress-versus-baseline AUC 0.952 against 0.951) and behaves as an RC element with the same half-life. The measured stretch never improves on that element, and two-composition band-passes fall below the phasic band.

Behavioural models fitted to the measurements of chapter 4 also support reservoir simulations with a linear read-out. Each node follows its measured relaxation kernel and is bounded by the measured potentiation peak and pulse count. Under this model, the heterogeneous bank raises total linear memory from 3.22 to 4.44 (10/10 seeds) and mean information-processing capacity from 5.18 to 9.77. On WESAD, a lead-cell reservoir scores 0.671 on windowed binary EDA, below a static control (0.714). Streaming three-class macro-F1 is 0.562 for both heterogeneous and homogeneous banks, above linear EMA controls (0.508). Under synthetic corruption, the heterogeneous bank retains 0.811 while instantaneous input falls to 0.641, and a linear EMA bank at the same half-lives retains 0.808.

The additional value of a timescale distribution depends on the task and metric. It is large for random-input memory and nonlinear capacity, modest but subject-resolved for multi-lag physiological reconstruction ($+0.023$ R^2 over the slow homogeneous control), and absent for sustained WESAD labels. On reconstruction, a linear EMA bank at the same half-lives does better than any device bank (0.742 against 0.707), so the time constants themselves carry most of the benefit. Noise robustness replicates on the independent Non-EEG corpus and represents temporal filtering rather than a device-specific accuracy advantage.

The chapter therefore finds a practical role for the composition-set time constant as a designed filter element, and little added value from random banks over linear filters at the same time constants. The next experimental steps follow from that role: replaying recorded electrodermal signals

through a fabricated device, measuring the relaxation of a small write on a written state, and comparing a device with a printed RC element of the same time constant. A matched-amplitude pulse protocol and a replicated chemistry study would then test the model assumptions and extend the range of available time constants.

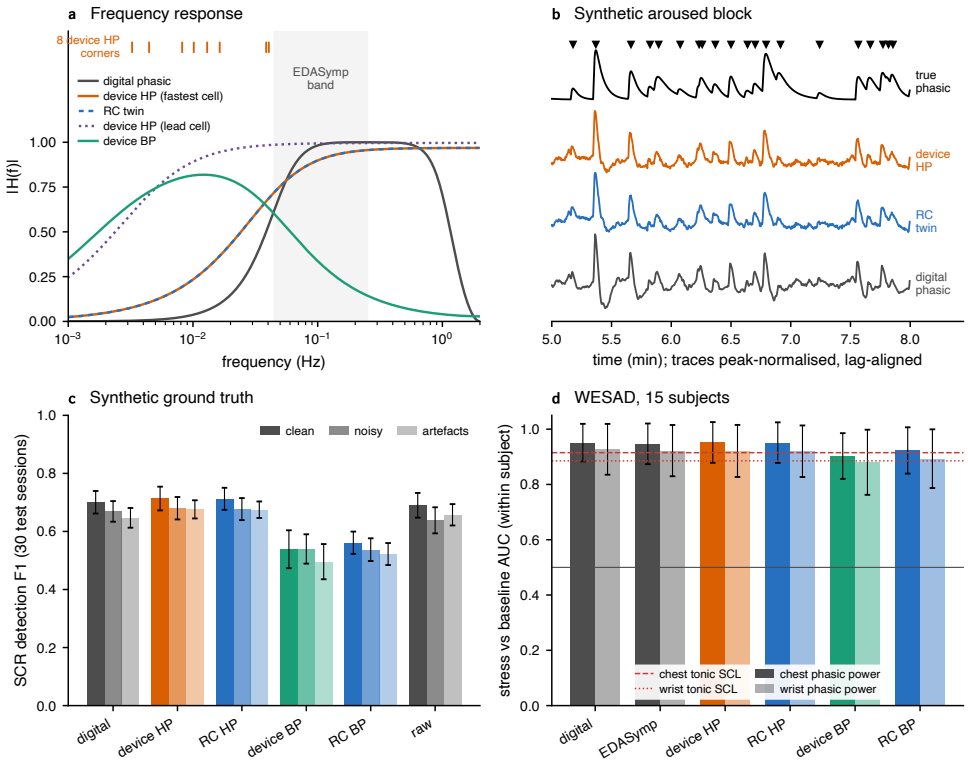


Figure 5.1: Composition-selected devices used as designed filters for electrodermal activity (simulations of the measured kernels). (a) Magnitude responses of a digital phasic filter, a single-device high-pass built from the fastest cell, the *RC* element with the same half-life, a high-pass built from the lead cell, and the best two-composition band-pass; ticks mark the high-pass corners of all eight cells, all below the sympathetic EDASymp band [1]. (b) Lag-aligned outputs during a synthetic aroused block; triangles mark the true SCR peaks. (c) SCR detection F1 on 30 synthetic test sessions (mean ± SD) for clean, noisy, and artefact-laden signals, with thresholds calibrated on 12 separate sessions. (d) Within-subject stress-versus-baseline AUC of per-minute phasic power on WESAD chest and wrist recordings (15 subjects, mean ± SD); dashed lines give the AUC of the tonic level alone. Device instances carry the measured within-cell timescale scatter; *RC* elements are ideal.

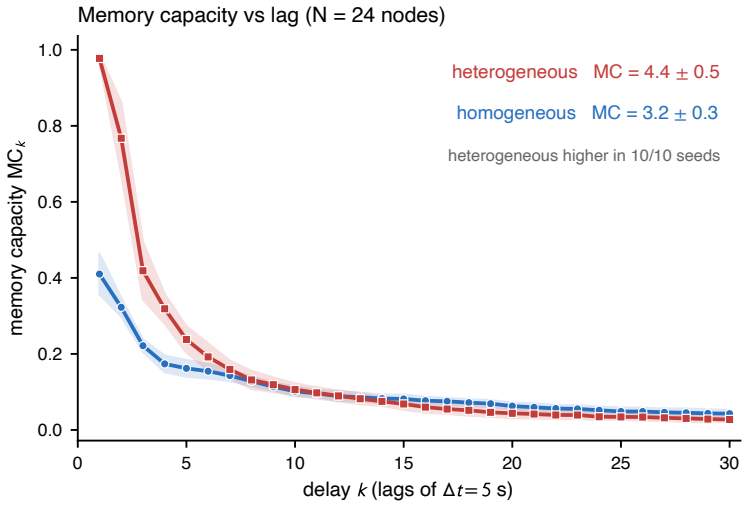


Figure 5.2: Linear memory capacity MC_k as a function of delay k for a homogeneous bank (all PEO 0.3/0.09 nodes) versus a heterogeneous composition bank ($\tau \approx 4$–$30\,\mathrm{s}$). Total capacity is 3.22 ± 0.32 and 4.44 ± 0.45, respectively; the heterogeneous value is higher in all ten seed pairs, mostly through recall at the shortest lags. Shaded bands are $\pm 1\,\mathrm{SD}$ over ten random input/bank seeds. Seeds quantify algorithmic sensitivity rather than experimental replication. $N = 24$ independent leaky nodes, random uniform input, linear ridge read-out.

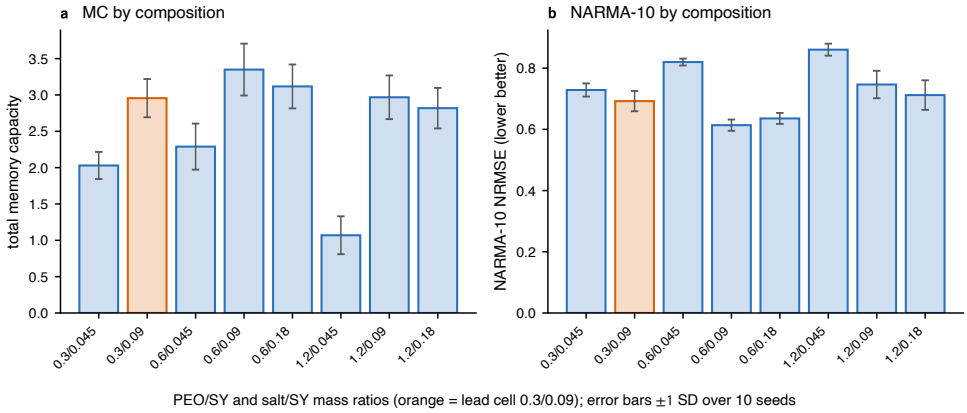


Figure 5.3: (a) Total linear memory capacity and (b) NARMA-10 reconstruction error (NRMSE, lower is better) for single-composition reservoirs across the PEO/LiOTf grid. The stretched PEO 0.6/0.09 cell gives the highest memory capacity and the lowest NARMA-10 error; the lead cell PEO 0.3/0.09 is shown in orange for reference. Only the *relative* composition ranking is claimed: absolute NARMA-10 NRMSE is high because a bank of independent, non-recurrent leaky nodes is a weak NARMA-10 reservoir.

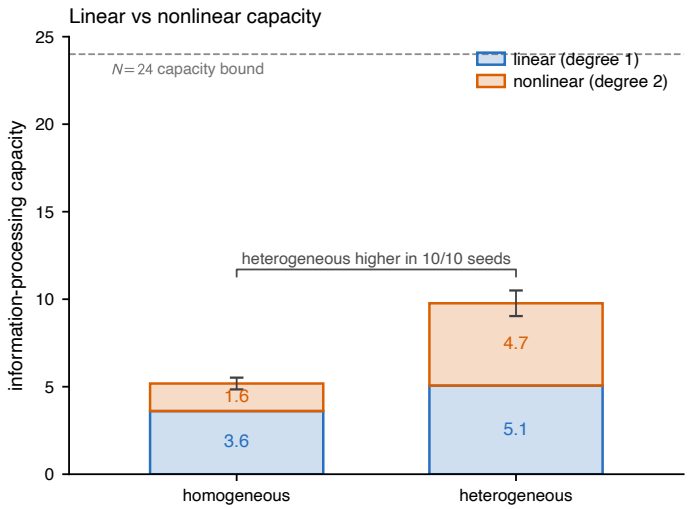


Figure 5.4: Information-processing capacity [16] split into linear (degree-one) and nonlinear (degree-two) parts for homogeneous and heterogeneous banks ($N = 24$, ten seeds). Both banks have nonzero degree-two capacity under the bounded node model. Mean total capacity changes from 5.18 ± 0.34 to 9.77 ± 0.73; the nonlinear component changes from 1.57 to 4.71 and is higher in all ten seed pairs. The dashed line is the $N = 24$ upper bound; seeds quantify algorithmic sensitivity rather than experimental replication.

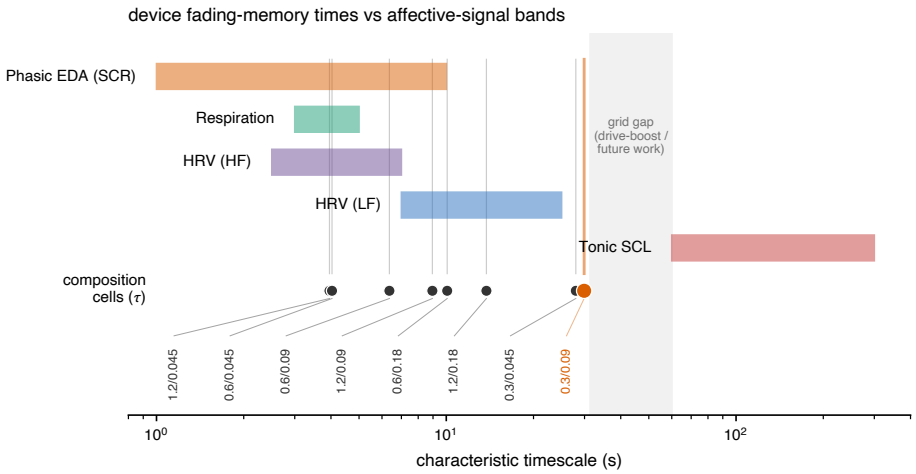


Figure 5.5: Each replicated PEO/LiOTf composition cell is placed at its derived relaxation time τ (markers, lower track) on the same logarithmic axis as characteristic timescale bands of peripheral affective signals (coloured bars; table 5.2). The grid ($\tau \approx 4$–$30\,\mathrm{s}$, anchored to measured $t_{1/2}$) overlaps the phasic-EDA, respiration, and HRV bands, with the lead cell PEO 0.3/0.09 (orange) near the slow LF-HRV band. Composition alone does not reach the tonic, minutes-scale band (shaded gap); testing that range would require a measured slower cell or matched drive-amplitude experiments.

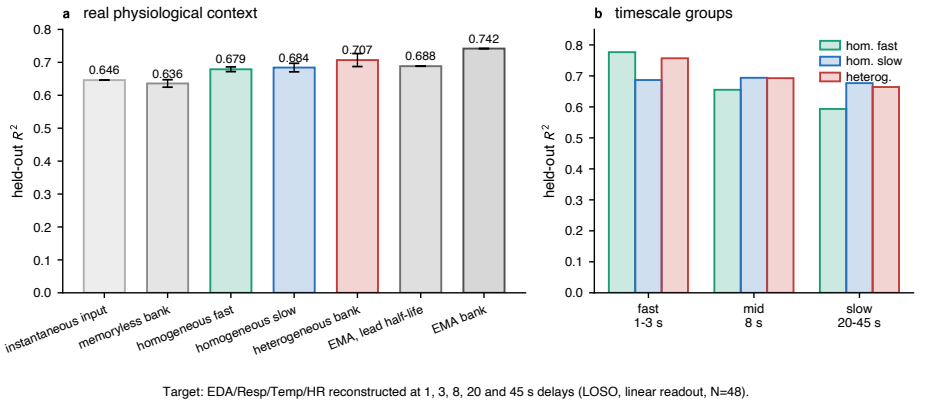


Figure 5.6: Multi-lag physiological context reconstruction on real WESAD streams (chest EDA, respiration, temperature, and causally derived heart rate). A linear read-out reconstructs each channel at delays of 1, 3, 8, 20 and 45 s (leave-one-subject-out, $N = 48$). Among device banks, the heterogeneous bank gives the highest overall subject-mean R^2 (0.707), ahead of the slow homogeneous control (0.684) and instantaneous input (0.646); a linear EMA bank at the same eight half-lives reaches 0.742. The zero-based axes avoid magnifying these modest differences.

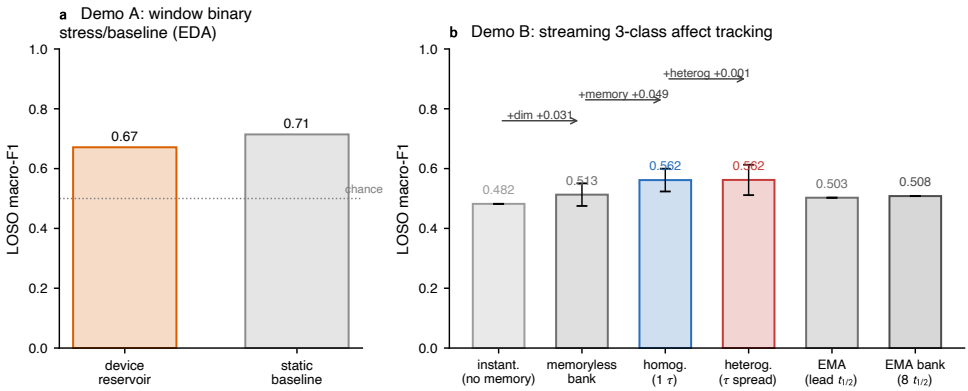


Figure 5.7: Affective classification with in-silico device reservoirs (leave-one-subject-out, linear read-out). (a) Window-level binary stress-versus-baseline from EDA: the device model scores 0.671, below the static-feature control (0.714). (b) Streaming three-class tracking. Fading memory improves on instantaneous input; the homogeneous and heterogeneous banks tie (0.562 ± 0.038 and 0.562 ± 0.051) and both exceed linear EMA controls (0.503 and 0.508). Error bars are ± 1 SD over ten bank seeds; both bar-chart axes begin at zero.

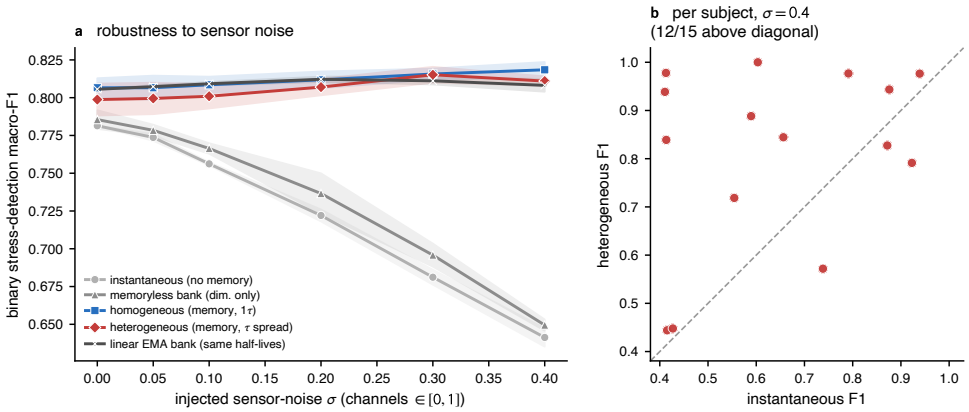


Figure 5.8: Continuous stress monitoring (binary stress vs. baseline-plus-amusement, per-step, leave-one-subject-out, linear read-out). (a) Detection macro-F1 against injected corruption. At $\sigma = 0.4$, instantaneous input falls to 0.641, while homogeneous and heterogeneous memory banks retain 0.818 and 0.811 and a linear EMA bank at the same half-lives retains 0.808; bands are ± 1 SD over seeds. (b) The heterogeneous bank exceeds instantaneous input in 12/15 subjects ($p = 0.015$). The homogeneous bank's score shows that timescale spread is not required.

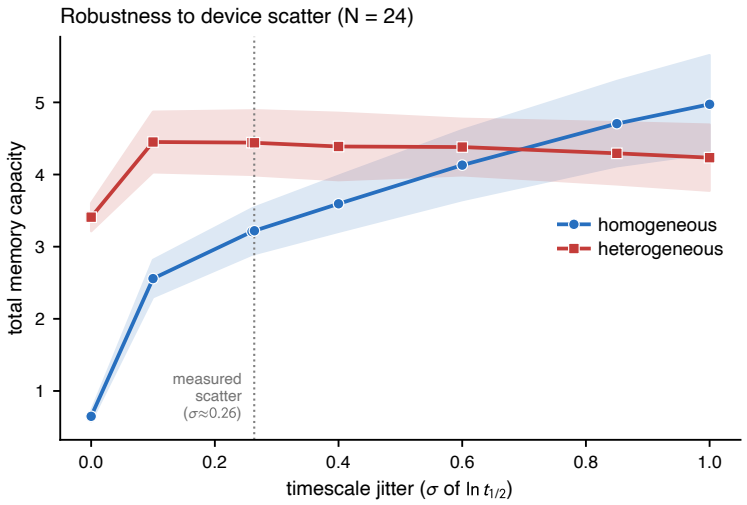


Figure 5.9: Total memory capacity versus injected log-normal timescale jitter for homogeneous and heterogeneous banks ($N = 24$, ten seeds; shaded bands ± 1 SD). The dotted line is the primary screened, substrate-level estimate $\sigma(\ln t_{1/2}) = 0.264$ (Appendix D). Only the timescale is perturbed; α and β retain their measured cell-card values. Larger jitter values, including the legacy fit-based value near 0.85, are stress tests. The curves describe algorithmic sensitivity and do not establish variability as beneficial.

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